Veriloga:

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// VerilogA for digital_logic, VendingMac, veriloga

`include "constants.vams"
`include "disciplines.vams"

module VendingMac(Q,D,R,Y,Q0,Q1,CLK);
inout Q,D,R,Y,Q0,Q1,CLK;
electrical Q,D,R,Y,Q0,Q1,CLK;


//internal nodes for FF inputs and return storage for interruptions
electrical T1, T0, R_S;

electrical Q0bar,Q1bar;

//parameters
parameter real vdd = 1.2 from (0:inf);
parameter real vth = 0.6 from (0:inf);
parameter real tr = 50p from (0:inf);
parameter real tf = 50p from (0:inf);

//boolean flags
real t1_b, t0_b, y_b, r_b,rs_b;

analog begin
t1_b=0;
t0_b=0;
@(cross(V(CLK) - vth, +1)) begin
r_b = 0;
y_b = 0;
end

if(((V(D)>vth)&&(V(R_S)<vth)&&(V(Q)<vth))||((V(D)<vth)&&(V(R_S)>vth)&&(V(Q)<vth))||((V(D)<
vth)&&(V(R_S)<vth)&&(V(Q)>vth)))begin

        //T0
        t0_b = 0;

if(((V(Q0)>vth)&&(V(R_S)>vth))||((V(Q1)>vth)&&(V(R_S)>vth))||((V(Q1)<vth)&&(V(Q)>vth)))
        t0_b = 1;
```

```
		//T1
		t1_b = 0;

if(((V(Q1)>vth)&&(V(Q)>vth))||((V(Q1)>vth)&&(V(R_S)>vth))||((V(Q0)>vth)&&(V(Q)>vth)))
		t1_b = 1;
end

//updates on clock rising
@(cross(V(CLK) - vth, +1)) begin
		//make sure that oinly valid inputs are present: only one thing can happen at a time
if(((V(D)>vth)&&(V(R_S)<vth)&&(V(Q)<vth))||((V(D)<vth)&&(V(R_S)>vth)&&(V(Q)<vth))||((V(D)<
vth)&&(V(R_S)<vth)&&(V(Q)>vth)))begin
		//Y
		y_b = 0;
		if((V(D)>vth)||((V(Q1)>vth)&&(V(Q)>vth)))
		y_b = 1;

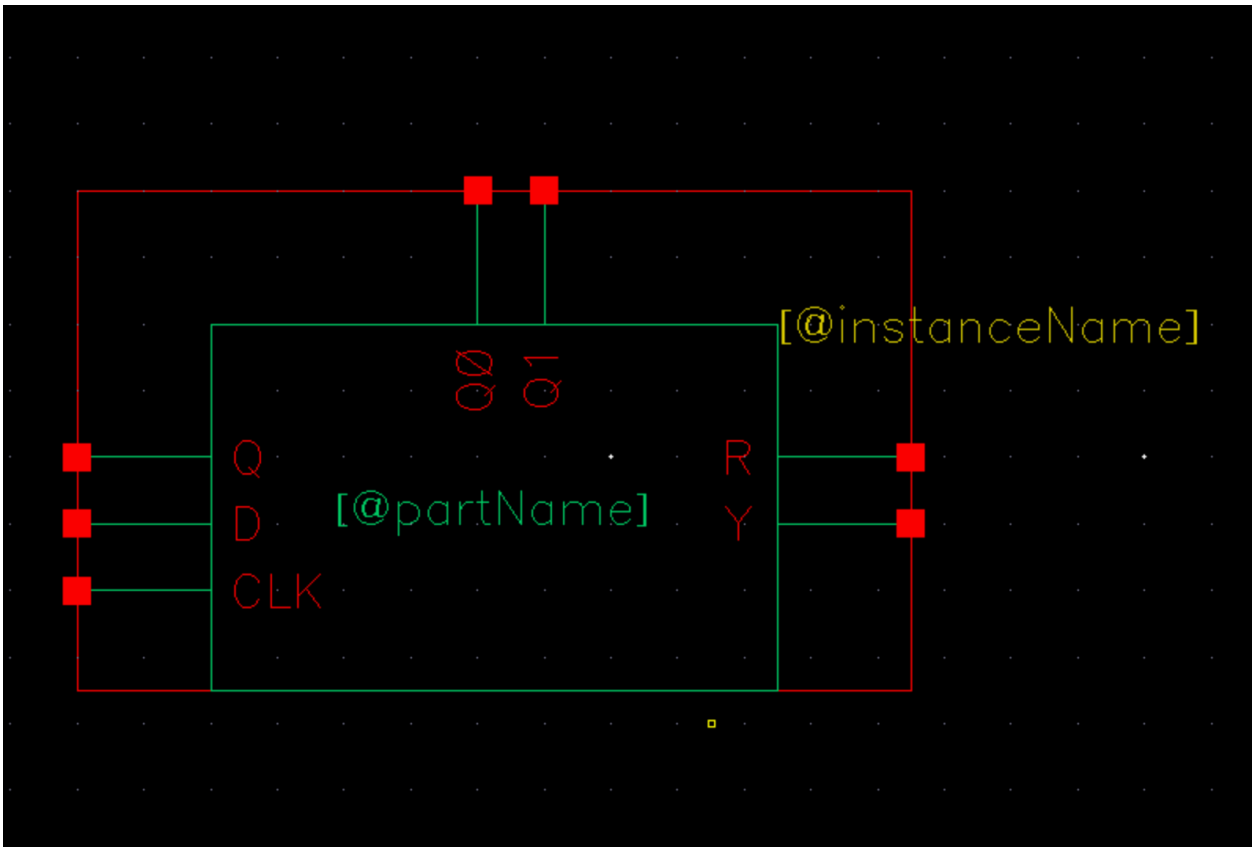
		//R
		r_b = 0;
		rs_b = 0;
		if((V(D)>vth)||((V(Q1)>vth)&&(V(R_S)>vth))||((V(Q0)>vth)&&(V(R_S)>vth)))
		r_b = 1;
		rs_b = 1;

		end

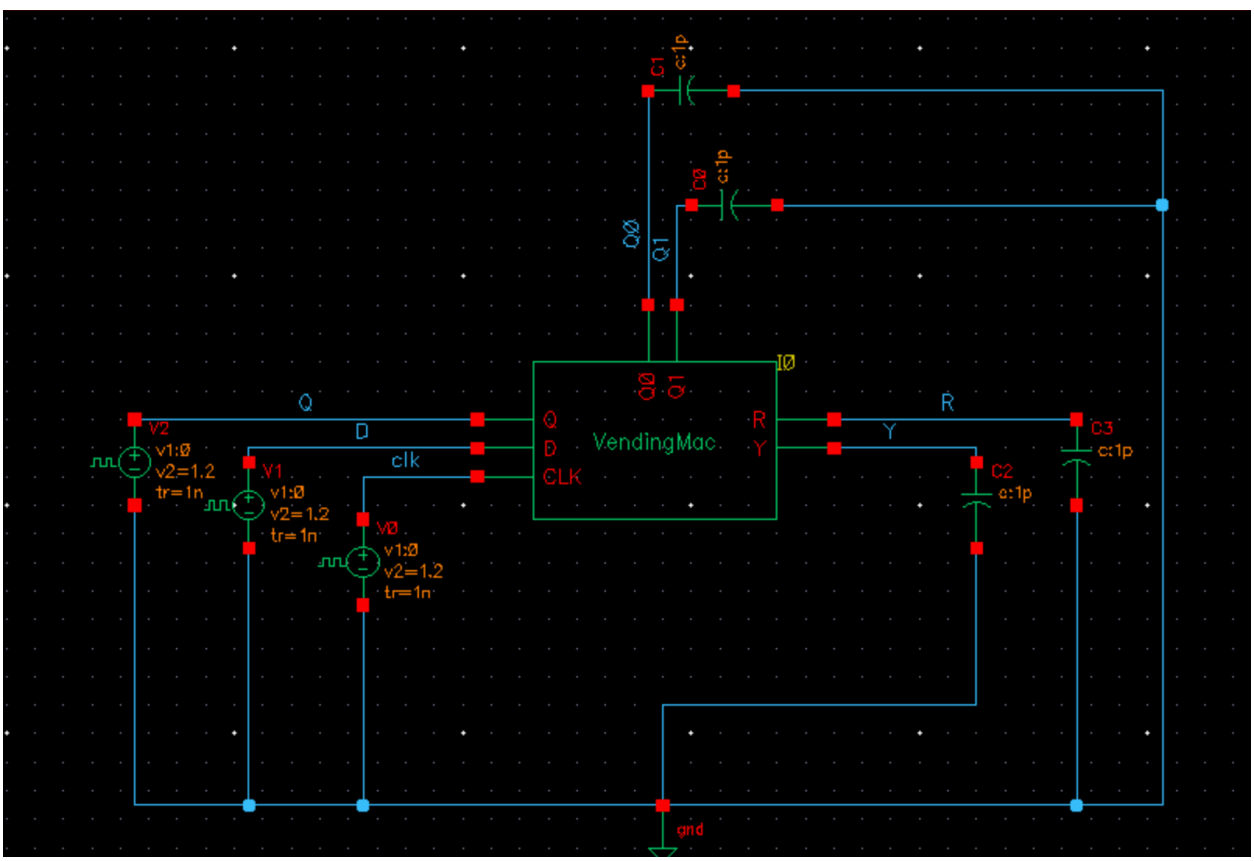
end
//drive electrical nodes
V(T1) <+ transition(vdd*t1_b, 0, tr, tf);
V(T0) <+ transition(vdd*t0_b, 0, tr, tf);
V(Y) <+ transition(vdd*y_b , 0, tr, tf);
V(R) <+ transition(vdd*r_b , 0, tr, tf);
V(R_S) <+ transition(vdd*r_b , 0, tr, tf);
end

T_FLIPFLOP FF0(.Q(Q0), .T(T0), .CLK(CLK), .Qbar(Q0bar));
T_FLIPFLOP FF1(.Q(Q1), .T(T1), .CLK(CLK), .Qbar(Q1bar));

endmodule
```


[@instanceName]
Q0
Q1
Q
D
CLK
[@partName]
R
Y

Test bench


C1
c:1p
C0
c:1p
Q0
Q1
I0
Q0
Q1
Q
D
CLK
VendingMac
R
Y
R
Y
Q
D
clk
V2
v1:0
v2=1.2
tr=1n
V1
v1:0
v2=1.2
tr=1n
V0
v1:0
v2=1.2
tr=1n
C3
c:1p
C2
c:1p
gnd

Testing:

All cases will be using:

<u>CLK (V0):</u>

- Voltage 1: 0V
- Voltage 2: 1.2
- Period: 1u
- Delay Time: 0
- Rise Time: 1n
- Fall Time: 1n
- PW: 500ns

Transaction Scenario #1: 3 quarters (20u)

Transaction Scenario #2: 2 quarters and 1 dollar (20u)

Transaction Scenario #1: 1 dollar (20u)

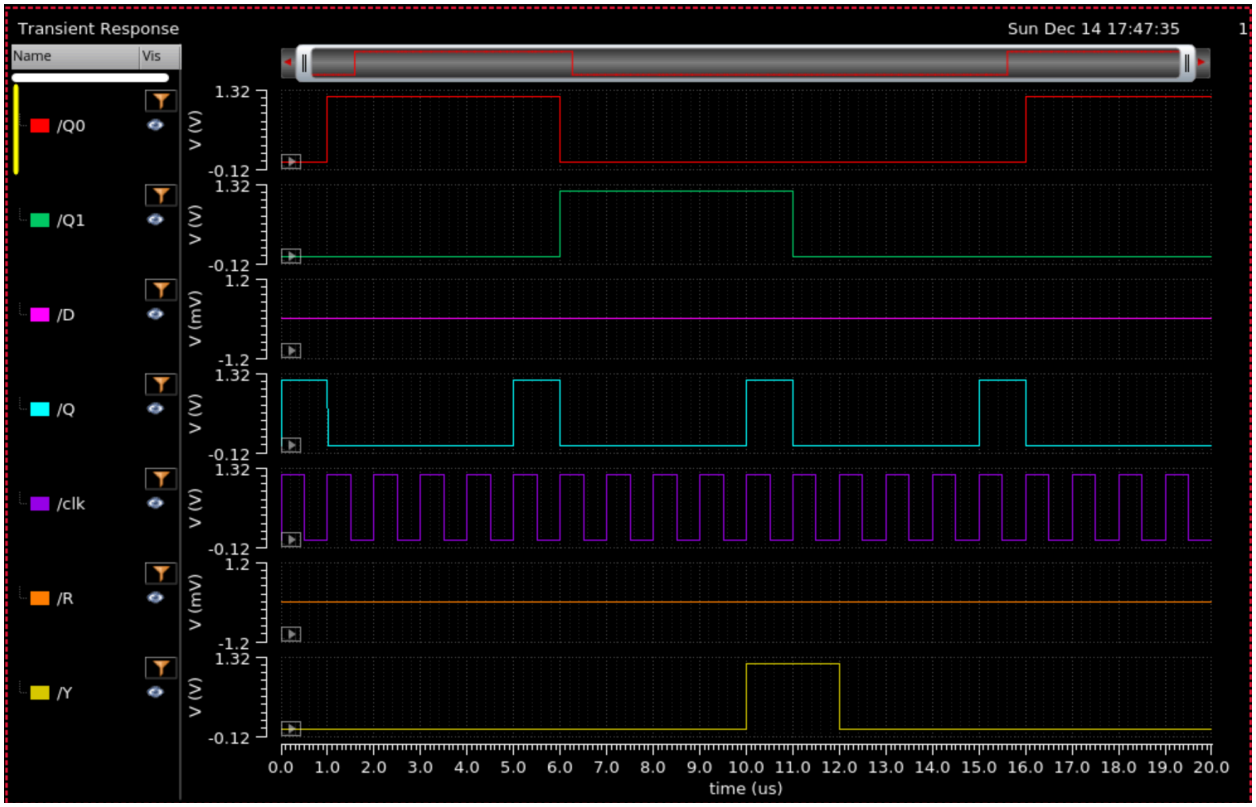
Transaction Scenario #1: 3 quarters (20u)

<u>D (V1):</u>

- Voltage 1: 0V
- Voltage 2: 1.2
- Period: 3u
- Delay Time: 1u
- Rise Time: 1n
- Fall Time: 1n
- PW: 750ns

<u>Q (V2):</u>

- All Default 0



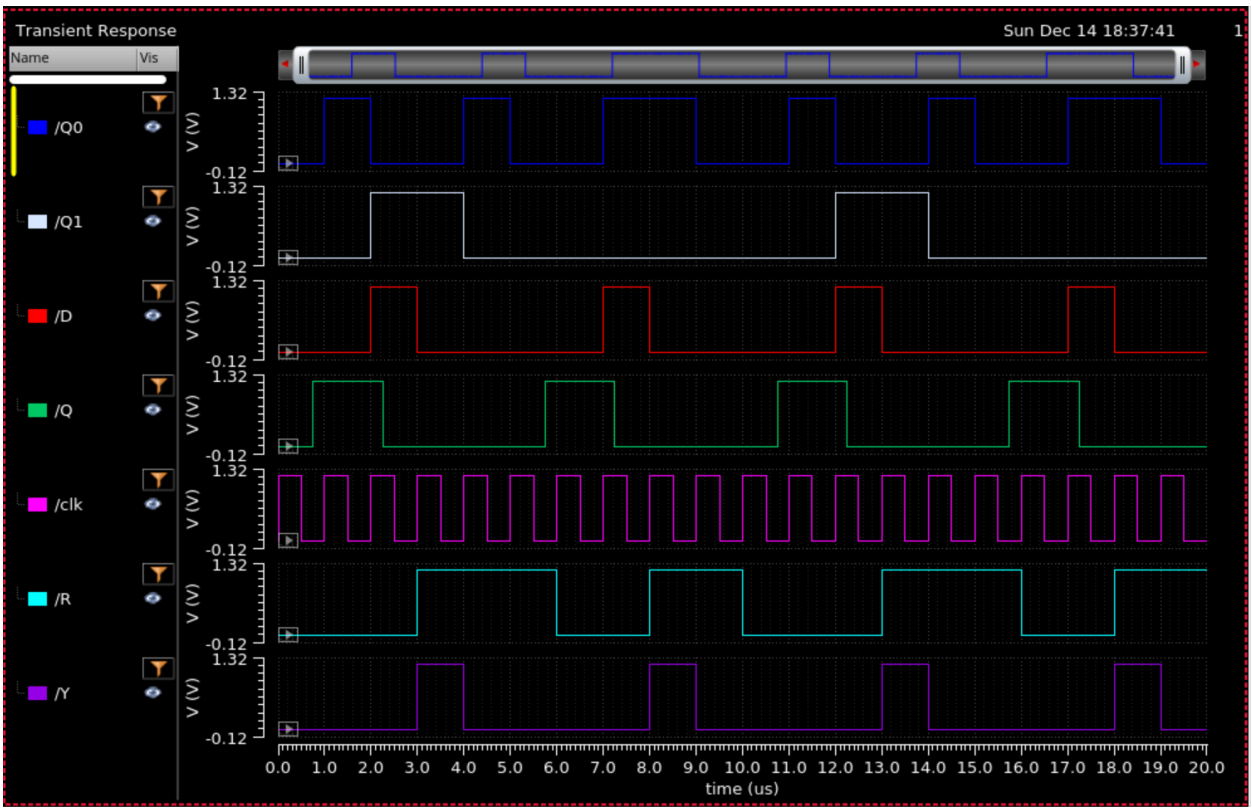
Transaction Scenario #2: 2 quarters and 1 dollar (20u)

<u>D (V1):</u>

- Voltage 1: 0V
- Voltage 2: 1.2
- Period: 5u
- Delay Time: 2.25u
- Rise Time: 1n
- Fall Time: 1n
- PW: 1u

<u>Q (V2):</u>

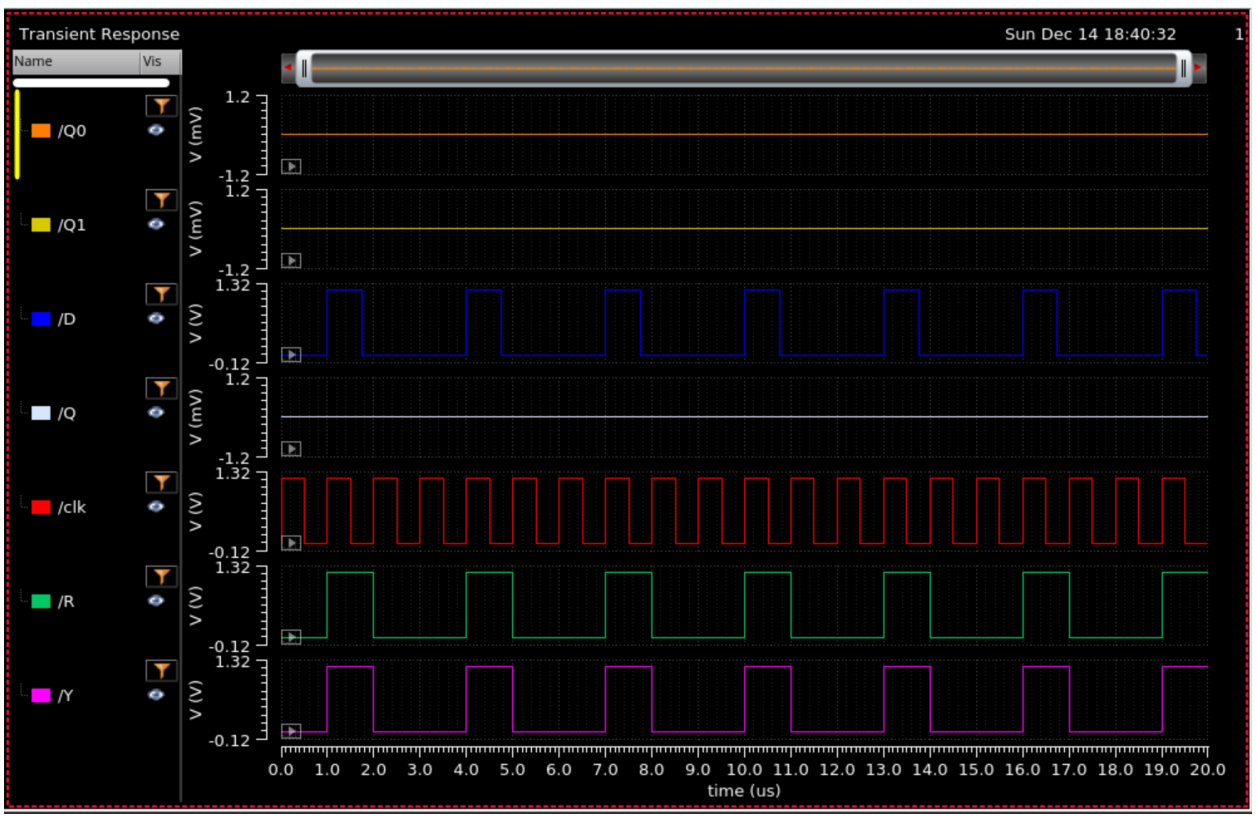
- Voltage 1: 0V
- Voltage 2: 1.2
- Period: 5u
- Delay Time: 750ns
- Rise Time: 1n
- Fall Time: 1n
- PW: 1.5u



Transaction Scenario #3: 1 dollar (20u)

<u>D (V1):</u>

- Voltage 1: 0V
- Voltage 2: 1.2
- Period: 1u
- Delay Time: 1u
- Rise Time: 1n
- Fall Time: 1n
- PW: 750n

<u>Q (V2):</u>

- Default 0